

CSD 2258
DATA MINING TOOLS LABORATORY

**Indian Startup Funding Analysis and
Funding Stage Prediction using Data Mining Techniques.**

MEMBERS:

SYED RUKNUDEEN AFZAL – 240171601062

MD WASEEUDDIN M – 240171601023

ABSTRACT:

This project focuses on analyzing the Indian Startup Funding ecosystem using the Indian Startup Funding dataset (2020–2025). The objective is to apply data preprocessing, classification, clustering, and association rule mining techniques to extract meaningful insights. Preprocessing steps such as handling missing values, encoding categorical variables, and feature scaling were performed. Logistic Regression and Decision Tree classifiers were used for prediction, along with a Random Forest model. The Apriori algorithm was applied for association analysis of industry and city funding patterns, and K-Means clustering was used for segmentation of startups based on funding amount. The models were evaluated using accuracy, precision, recall, and confusion matrix.

KEYWORDS

Startup Funding, Logistic Regression, Decision Tree, Random Forest, Apriori Algorithm, K-Means Clustering, Data Mining, Indian Startup Ecosystem

INTRODUCTION

The Indian startup ecosystem has grown significantly between 2020 and 2025, attracting investments across diverse industries such as Fintech, EdTech, HealthTech, and Mobility. Understanding funding patterns is crucial for entrepreneurs, investors, and policymakers. By analyzing transaction-level funding data, stakeholders can identify which industries receive the most investment, which cities are startup hubs, and what factors determine whether a startup reaches late-stage funding. In this project, machine learning and data mining techniques are applied to the Indian Startup Funding dataset to classify funding stages, discover industry-city funding associations, and segment startups into meaningful groups based on investment size.

LITERATURE REVIEW

Previous studies in startup analytics and venture capital prediction focus on identifying factors that drive funding success. Logistic Regression and Decision Trees are commonly used for binary classification tasks such as predicting whether a startup will receive follow-on funding. Random Forest, as an ensemble method, has shown superior performance in handling imbalanced and high-dimensional data. Apriori-based association rule mining has been applied in business analytics to find co-occurrence patterns in categorical data such as industry-city combinations. Clustering techniques such as K-Means help segment startup ecosystems by funding level for targeted strategy development.

METHODOLOGY:

Dataset:

- Source: Indian Startup Funding Dataset (Kaggle – Vagdevi Titiksha G)

- File size: 101.2 kB
- File name: indian_startup_funding_2020_2025.csv
- Target Variable: Funding Stage (Early / Late)

Key Features:

- Startup Name
- Industry
- SubVertical
- City
- Investors
- Investment Type
- Investment Amount
- Date

	Startup	Industry	SubVertical	City	Investors	Investment Type	InvestmentAmount_USD	Date
1	Housejoy	EdTech	K12	Mumbai	Lightspeed India	Seed	199000	19-04-2023
2	Groww	Media	Streaming	Bengaluru	IFC	Seed	1668000	28-01-2025
3	Groww	Mobility	Ride Sharing	Hyderabad	Nexus Venture Partners, Peak XV	Series B	38052000	14-03-2021
4	FarmBox	Consumer Electronics	Wearables	Gurugram	Kalaari Capital, Y Combinator	Seed	455000	11-09-2023
5	Udaan	RealEstate	Rental Tech	Mumbai	Bessemer Venture Partners	Seed	89000	31-01-2024
6	AeroStack	EdTech	Coding Bootcamp	Delhi	Matrix Partners India, Peak XV	Pre-Series A	143000	24-07-2021
7	Ola	Retail	Fashion	Mumbai	Tiger Global Management	Seed	62000	12-01-2023
8	FinSpace	Consumer Electronics	Wearables	Gurugram	Blume Ventures	Growth	491721000	22-01-2025
9	Rapido	FoodTech	Food Delivery	Kolkata	A91 Partners, Kedaara Capital, Tiger Global Manage...	Seed	459000	27-11-2023
10	HyperLoop	EdTech	Test Prep	Ahmedabad	Mirae Asset, SoftBank Vision Fund, Tiger Global	Series B	35610000	04-05-2020
11	NextSense_10	Mobility	Ride Sharing	Delhi	IFC, Kalaari Capital, Peak XV	Seed	54000	19-05-2020
12	LogiMart	FinTech	NeoBank	Mumbai	Falcon Edge, Peak XV	Pre-Series A	933000	10-04-2023
13	CoreFlow	SaaS	DevTools	Hyderabad	Elevation Capital, SoftBank Vision Fund, Ventures In...	Pre-Series A	680000	07-10-2023
14	Groww	FinTech	InsurTech	Ahmedabad	Falcon Edge	Debt	374000	23-07-2020
15	ClearDekho	Logistics	Last Mile	Pune	Y Combinator	Private Equity	151993000	05-12-2024
16	NextSense	Mobility	Micro Mobility	Hyderabad	A91 Partners, Falcon Edge, Prosus Ventures	Series A	1643000	10-07-2024

DATA PREPROCESSING:

1. Handling Missing Values:

- Removed rows with missing Investment Amount
- Replaced missing Investors with "Unknown"
- Replaced missing SubVertical with "General"

2. Encoding Categorical Variables:

- Industry, City, and Investment Type converted to factors
- Target variable (Funding Stage) encoded as binary (0 = Early Stage, 1 = Late Stage)

3. Feature Scaling:

- Investment Amount standardized using `scale()` function in R

4. Data Splitting:

- Dataset split into 80% training and 20% testing using `createDataPartition()`

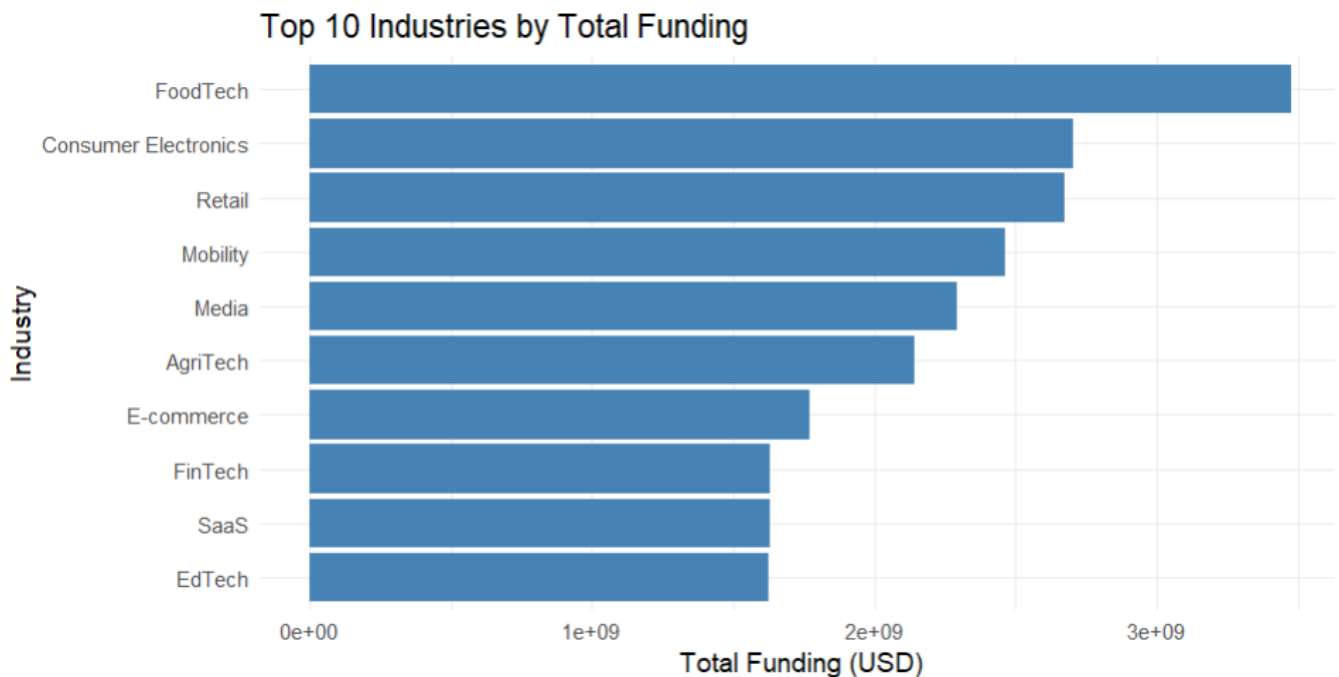
5. Target Variable Creation:

- Early Stage (0): Seed, Pre-Series A, Angel, Undisclosed
- Late Stage (1): Series A, Series B, Series C, Growth

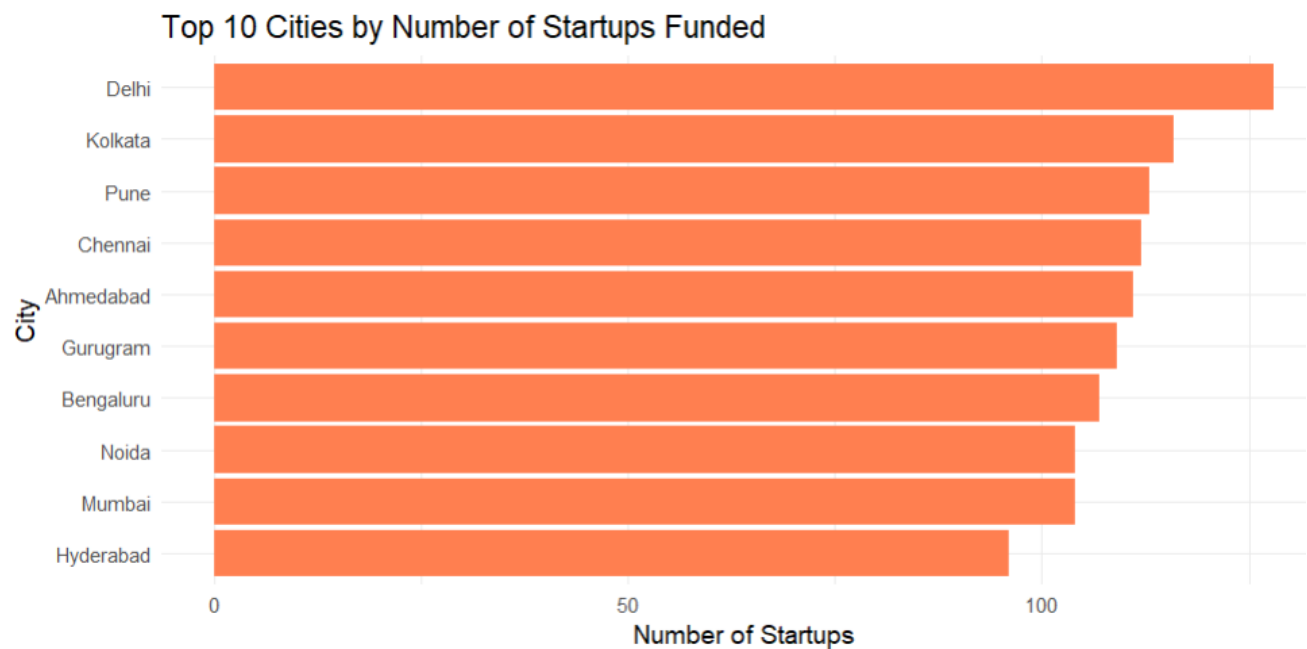
EXPLORATORY DATA ANALYSIS (EDA):

Before applying machine learning models, exploratory analysis was performed to understand the distribution and patterns in the dataset.

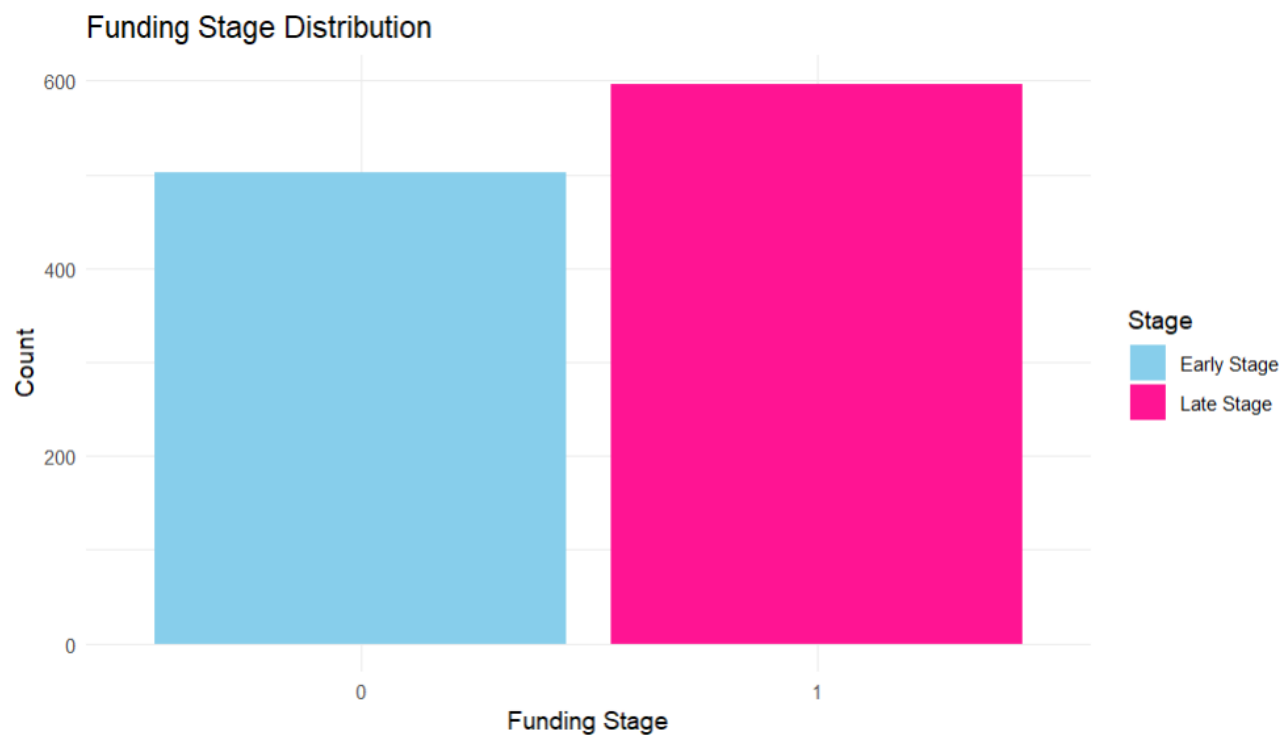
Top 10 Industries by Total Funding:



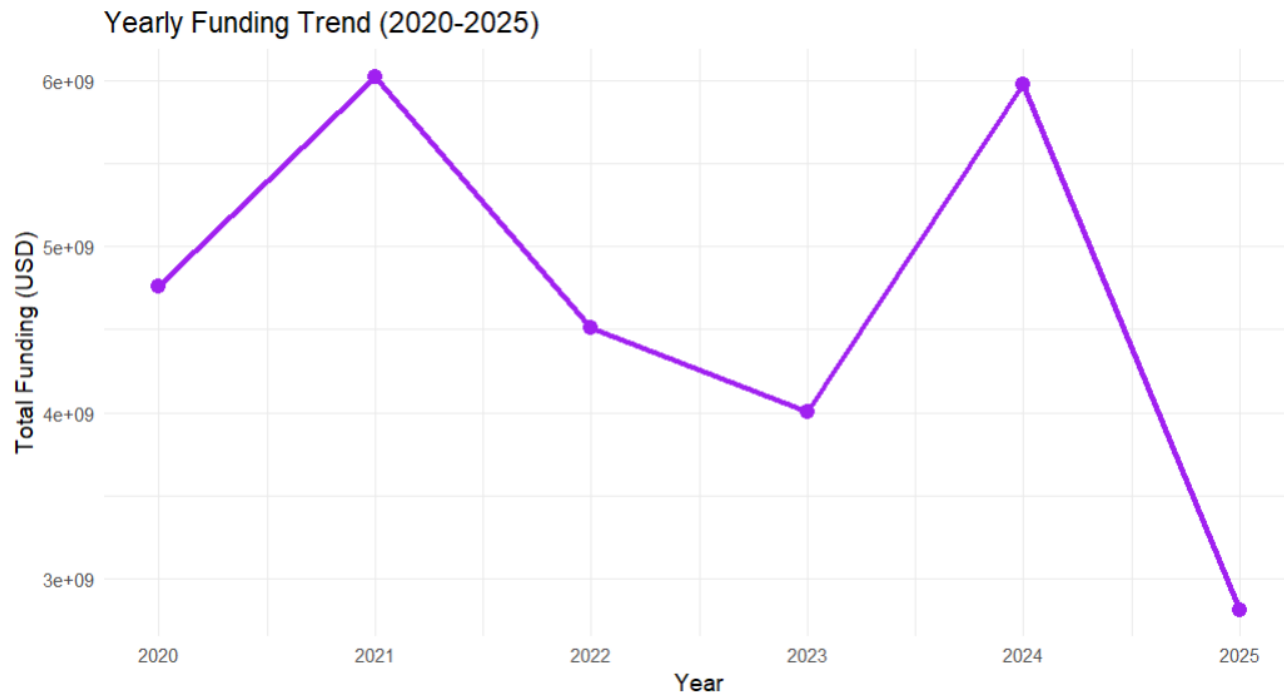
Top 10 Cities by Number of Startups Funded:



Funding Stage Distribution:



Yearly Funding Trend (2020–2025):



MODELS USED:

Logistic Regression:

Logistic Regression is a supervised machine learning algorithm used for binary classification problems. In this project, it is used to classify startups as Early Stage or Late Stage funders based on features such as Industry, City, Investment Amount, and Year.

The logistic regression curve is an S-shaped (sigmoid) curve that represents how the probability of the target variable changes with respect to an independent variable.

- The X-axis represents the feature (e.g., Investment Amount)
- The Y-axis represents the probability (0 to 1)
- The curve gradually increases and saturates at 1

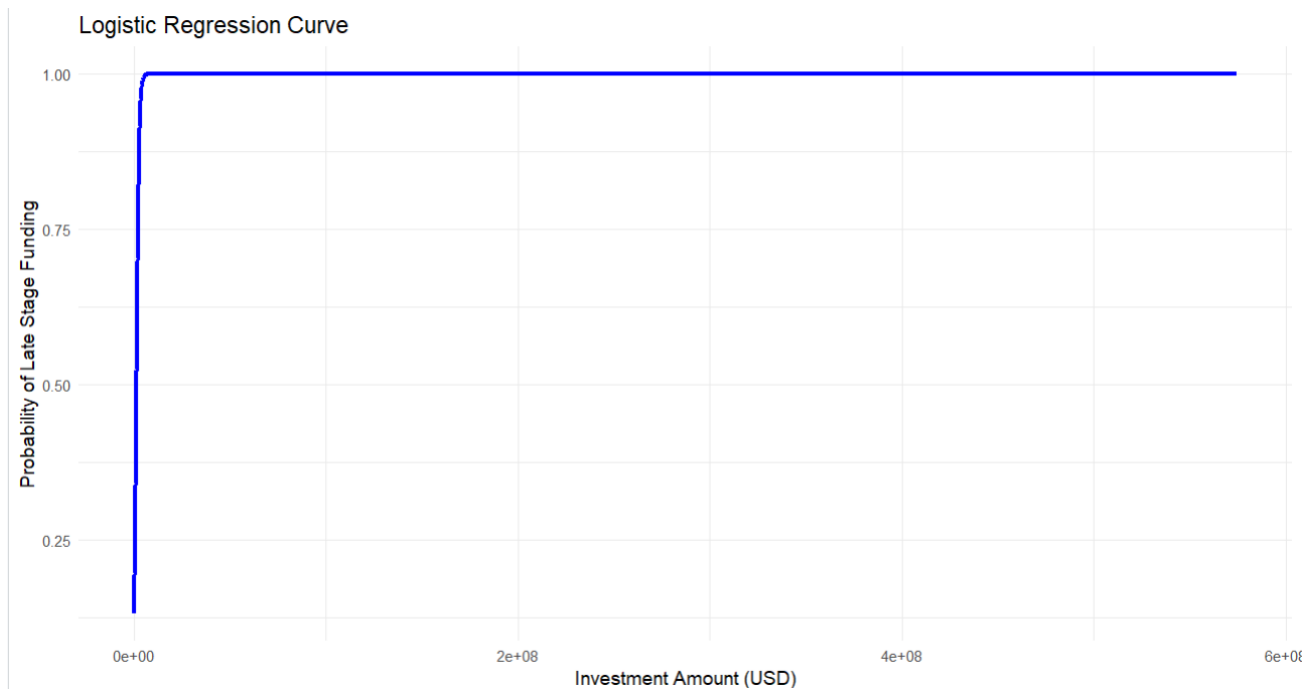
At a threshold value (commonly 0.5):

- If probability $\geq 0.5 \rightarrow$ classified as Late Stage (1)
- If probability $< 0.5 \rightarrow$ classified as Early Stage (0)

In the Indian Startup Funding dataset:

- As Investment Amount increases, the probability of reaching Late Stage funding increases
- The curve shows a smooth transition rather than a sharp boundary

- This helps in understanding startup funding behavior probabilistically



R 4.5.2 · C:/Users/syed/Downloads/r dataset/ ↗

```
> conf_mat_log <- confusionMatrix(log_pred, testData$FundingStage)
> print(conf_mat_log)
```

Confusion Matrix and Statistics

	Reference	
Prediction	0	1
0	90	30
1	10	89

Accuracy : 0.8174
 95% CI : (0.7597, 0.8662)
 No Information Rate : 0.5434
 P-Value [Acc > NIR] : < 2.2e-16

Kappa : 0.6377

Mcnemar's Test P-Value : 0.002663

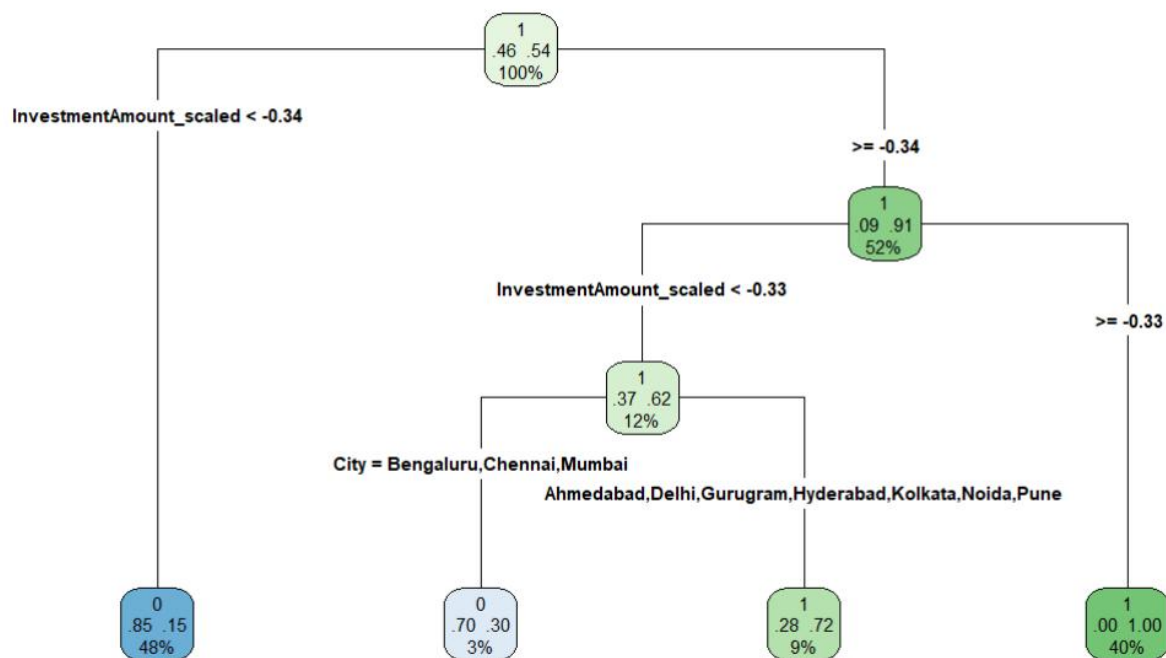
Sensitivity : 0.9000
 Specificity : 0.7479
 Pos Pred Value : 0.7500
 Neg Pred Value : 0.8990
 Prevalence : 0.4566
 Detection Rate : 0.4110
 Detection Prevalence : 0.5479
 Balanced Accuracy : 0.8239

'Positive' Class : 0

Decision Tree Classifier:

The Decision Tree model achieved an accuracy of approximately 84.01%, showing good classification performance on the startup funding dataset. Precision and recall values indicate balanced prediction capability across the classes. Investment Amount and City were among the important features influencing the predictions.

Decision Tree - Startup Funding Stage



R 4.5.2 · C:/Users/syed/Downloads/r dataset/

```
> cat("\nDecision Tree Accuracy:", conf_mat_dt$overall['Accuracy'], "\n")
```

Decision Tree Accuracy: 0.8401826

```
> cat("Decision Tree Precision:", conf_mat_dt$byClass['Precision'], "\n")
```

Decision Tree Precision: 0.7826087

```
> cat("Decision Tree Recall:", conf_mat_dt$byClass['Recall'], "\n")
```

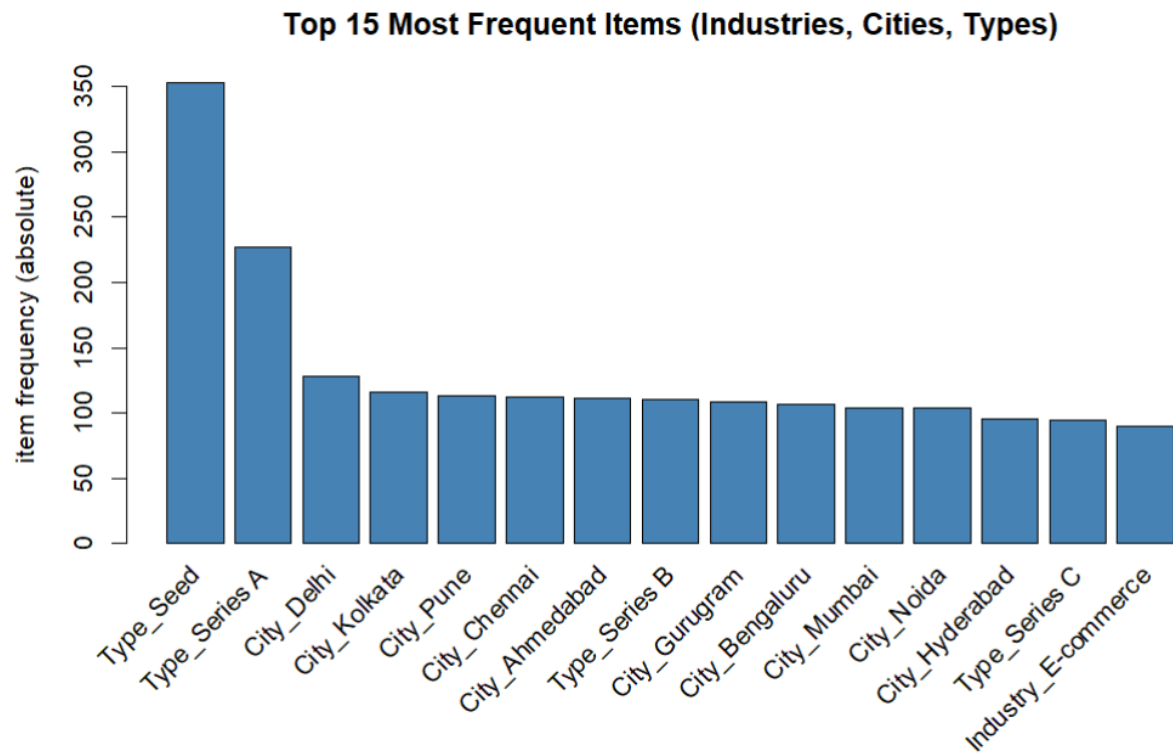
Decision Tree Recall: 0.9

```
> cat("Decision Tree F1:", conf_mat_dt$byClass['F1'], "\n")
```

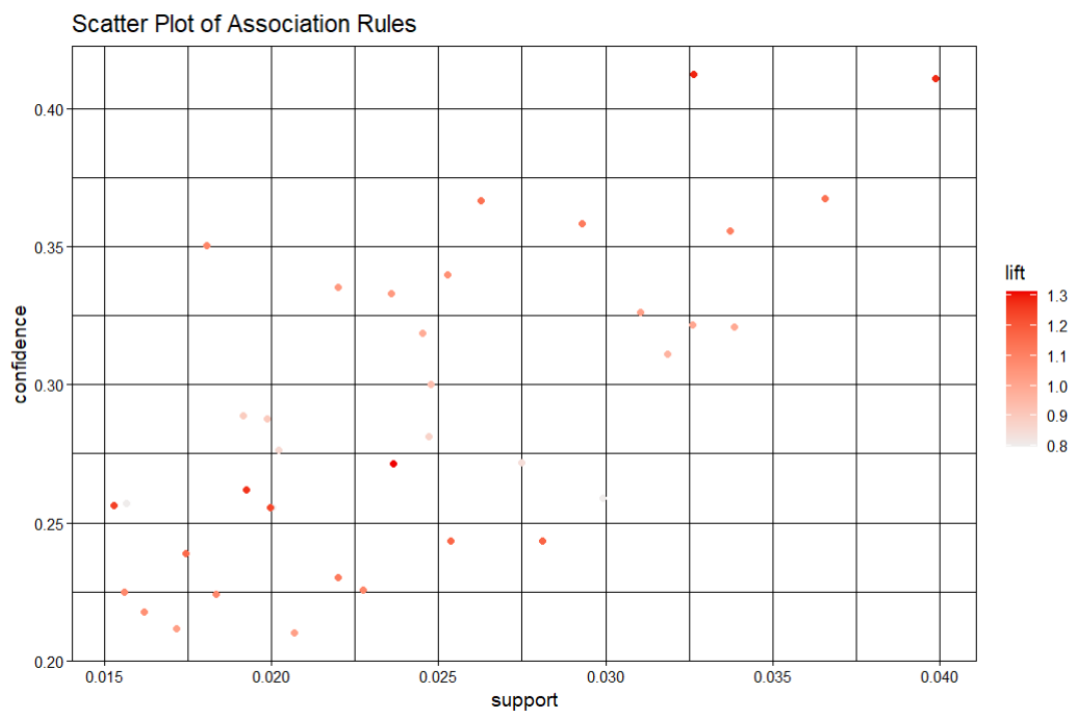
Decision Tree F1: 0.8372093

Association Rule Mining:

Association Rule Mining is a technique used to find relationships between items in large datasets. It is commonly used in market basket analysis to identify items that frequently occur together. In this project, the Indian Startup Funding dataset is converted into a transaction format, where each startup record represents a transaction and each Industry, City, and Investment Type label represents an item.



The scatter plot below represents the association rules generated using the Apriori algorithm. Color intensity shows lift strength, size represents support.

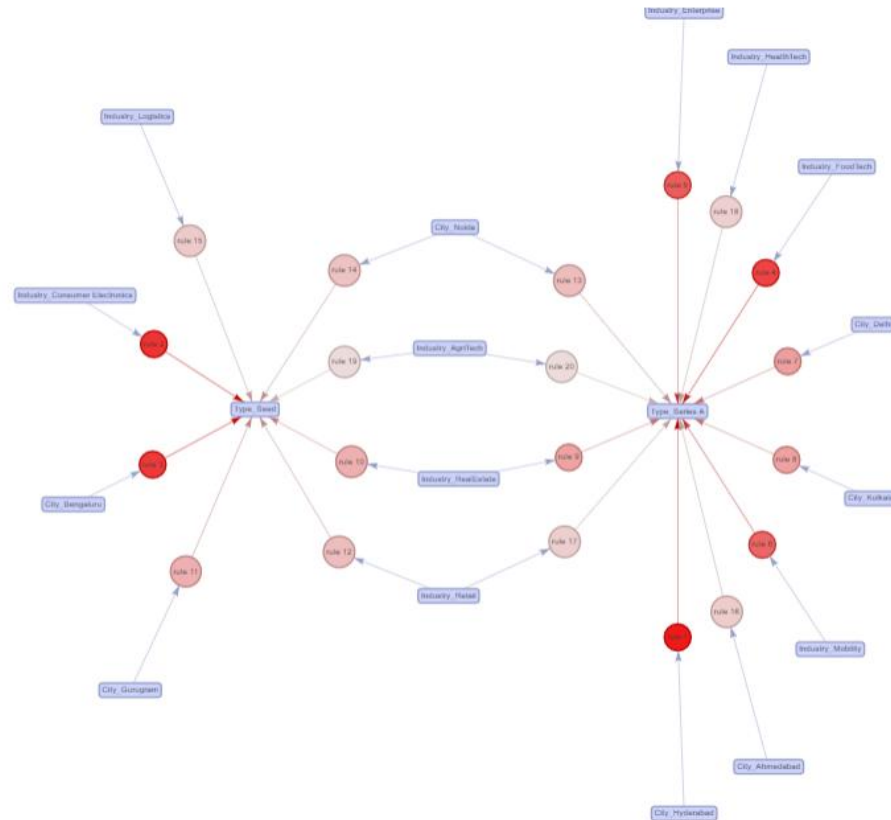


Frequent Itemsets

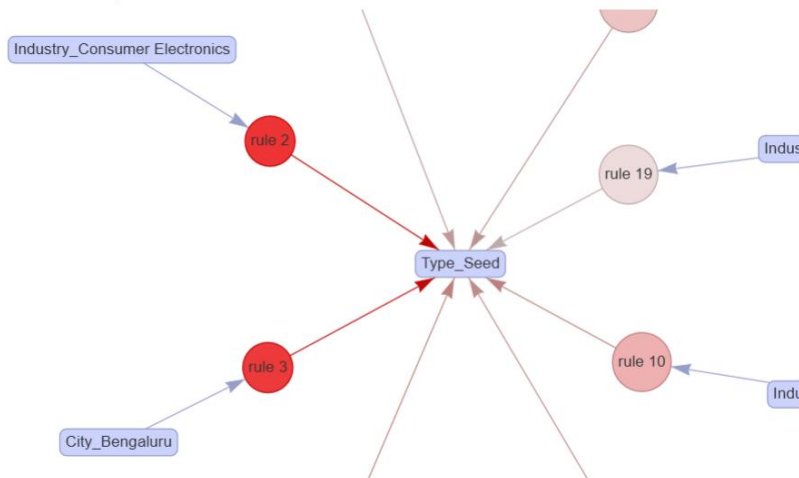
Frequent itemsets are groups of industry-city-funding type combinations that appear together in records. Examples from the dataset include:

- Industry_EdTech + City_Bengaluru
- Industry_Fintech + Type_Series A
- Industry_HealthTech + City_Mumbai

Select by id ▾



Select by id ▾



```
> inspect(head(rules_sorted, 10))
```

	lhs	rhs	support	confidence
[1]	{City_Hyderabad}	=> {Type_Series A}	0.02363636	0.2708333
[2]	{Industry_Consumer Electronics}	=> {Type_Seed}	0.03272727	0.4137931
[3]	{City_Bengaluru}	=> {Type_Seed}	0.04000000	0.4112150
[4]	{Industry_FoodTech}	=> {Type_Series A}	0.01909091	0.2625000
[5]	{Industry_Enterprise}	=> {Type_Series A}	0.01545455	0.2575758
[6]	{Industry_Mobility}	=> {Type_Series A}	0.02000000	0.2558140
[7]	{City_Delhi}	=> {Type_Series A}	0.02818182	0.2421875
[8]	{City_Kolkata}	=> {Type_Series A}	0.02545455	0.2413793
[9]	{Industry_RealEstate}	=> {Type_Series A}	0.01727273	0.2405063
[10]	{Industry_RealEstate}	=> {Type_Seed}	0.02636364	0.3670886

	coverage	lift	count
[1]	0.08727273	1.312408	26
[2]	0.07909091	1.289440	36
[3]	0.09727273	1.281406	44
[4]	0.07272727	1.272026	21
[5]	0.06000000	1.248164	17
[6]	0.07818182	1.239627	22
[7]	0.11636364	1.173596	31
[8]	0.10545455	1.169679	28
[9]	0.07181818	1.165449	19
[10]	0.07181818	1.143902	29

K-MEANS CLUSTERING:

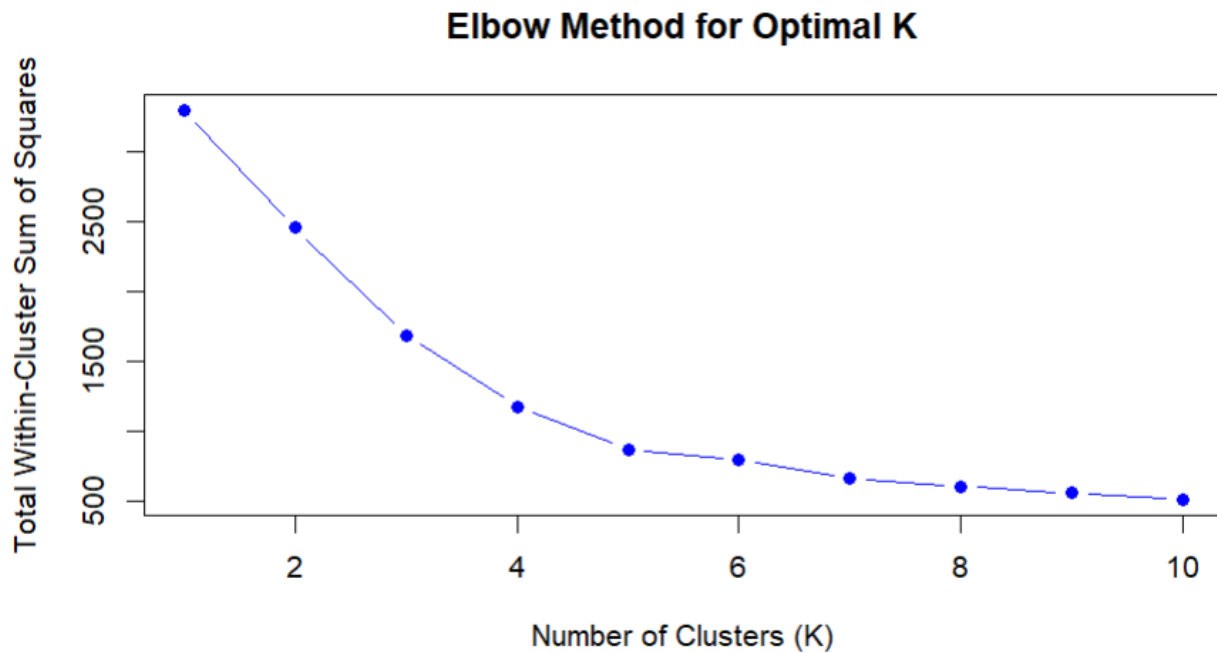
K-Means is an unsupervised machine learning algorithm used to group similar data points into clusters. It partitions the dataset into K distinct clusters, where each data point belongs to the cluster with the nearest mean (centroid).

In this project, K-Means is used to segment startups based on their funding behavior using features such as Investment Amount, Industry Code, and City Code.

The Elbow Method is used to determine the optimal number of clusters by plotting:

- X-axis → Number of clusters (K)
- Y-axis → Within-cluster sum of squares (WSS)

The elbow point indicates the best value of K where adding more clusters does not significantly reduce error.

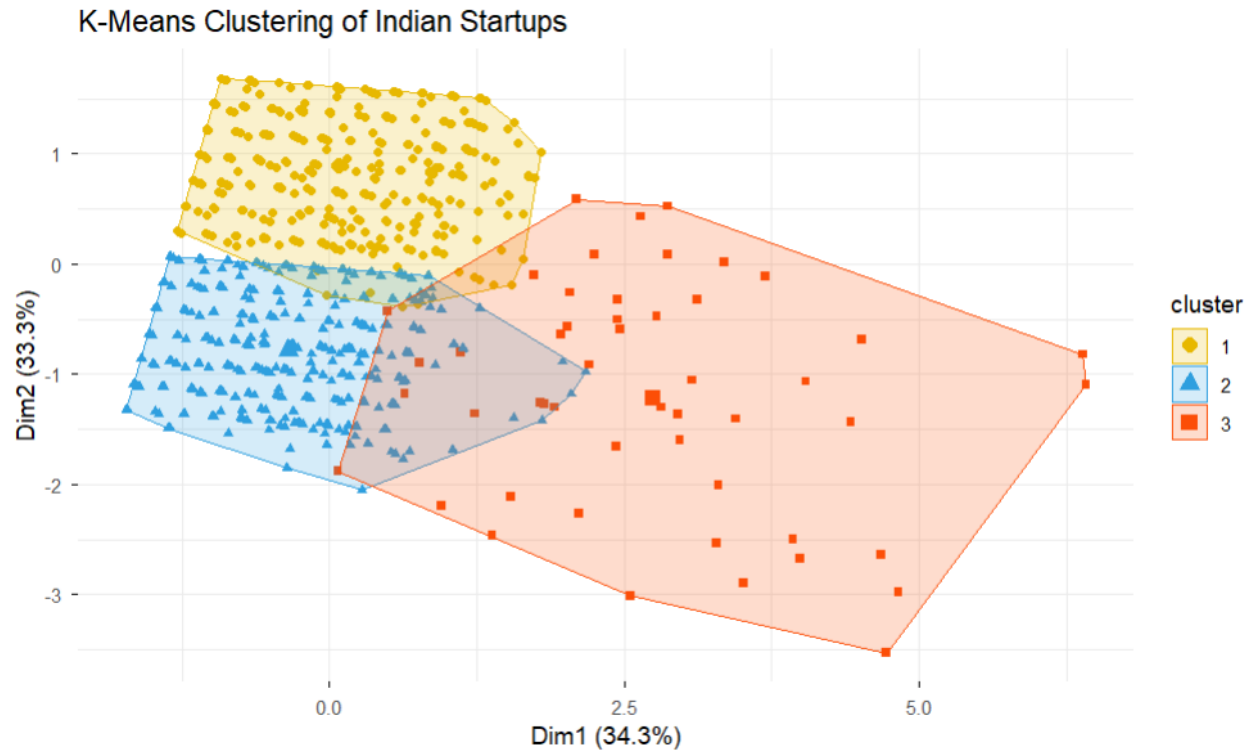


In the Indian Startup Funding dataset:

- Startups are grouped based on investment amount and industry patterns
- Features like Investment Amount help identify funding levels
- Clustering helps in understanding startup funding segments

APPLIED K-MEANS CLUSTERING:

Applied K-Means refers to the practical use of clustering results to derive meaningful business insights. After forming clusters, the results are analyzed to understand startup funding behavior and support investment decision-making.



After forming clusters, the results are analyzed to understand funding segments:

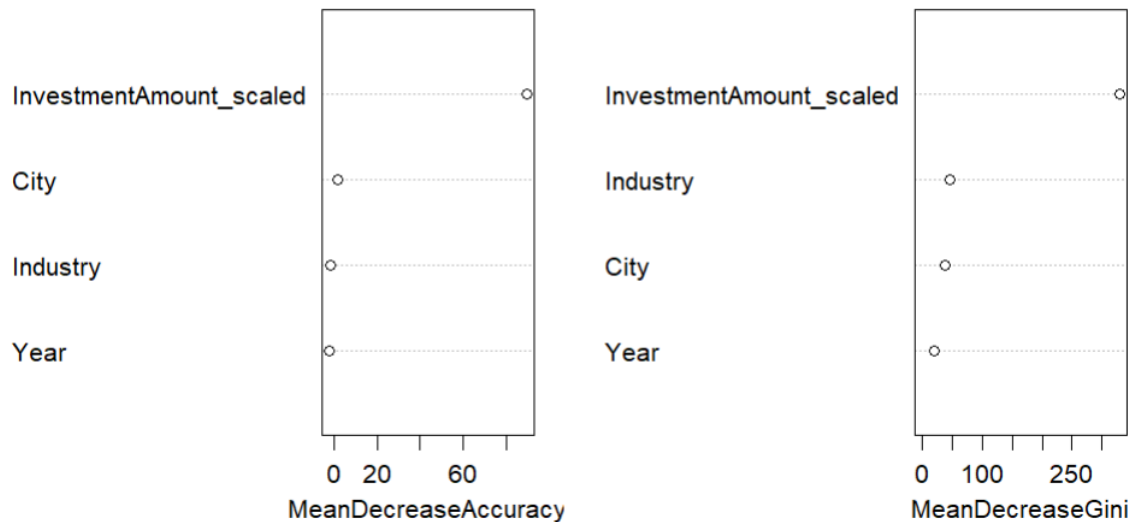
```
R • R 4.5.2 • C:/Users/syed/Downloads/r dataset/ ↗
Cluster Summary:
> print(cluster_summary)
# A tibble: 3 × 3
  Cluster Avg_InvestmentAmount Count
  <fct>      <dbl>     <int>
1 1        12190310.         529
2 2        11620461.         521
3 3        311666180          50
```

RANDOM FOREST CLASSIFIER:

Random Forest is an ensemble machine learning algorithm that builds multiple decision trees and combines their predictions using majority voting to improve accuracy and reduce overfitting. In this project, it is used to classify startups as Early Stage or Late Stage funders based on features like Investment Amount, Industry, City, and Year.

- The Random Forest model achieved an accuracy of approximately 81.27%.
- Precision, recall, and F1-score indicate balanced classification performance.
- Ensemble learning helped improve prediction stability and reduce overfitting compared to a single Decision Tree

Random Forest - Feature Importance



```
R 4.5.2 · C:/Users/syed/Downloads/r dataset/
> cat("\nRandom Forest Accuracy:", conf_mat_rf$overall['Accuracy'], "\n")
Random Forest Accuracy: 0.8127854
> cat("Random Forest Precision:", conf_mat_rf$byClass['Precision'], "\n")
Random Forest Precision: 0.7809524
> cat("Random Forest Recall:", conf_mat_rf$byClass['Recall'], "\n")
Random Forest Recall: 0.82
> cat("Random Forest F1:", conf_mat_rf$byClass['F1'], "\n")
Random Forest F1: 0.8
```

EVALUATION METRICS

To evaluate the performance of the classification models (Logistic Regression, Decision Tree, and Random Forest), several metrics were used. These metrics help measure how accurately the model predicts whether a startup is in Early Stage or Late Stage funding.

Performance Comparison:

<u>Model</u>	<u>Accuracy</u>	<u>Precision</u>	<u>Recall</u>	<u>F1</u>
Logistic Regression	81.73%	75%	90%	81.82%
Decision Tree	84.01%	78.26%	90%	83.72%
Random Forest	81.27%	78.09%	82%	80%

Insights

- Startups can be effectively classified into Early and Late funding stages using machine learning models, enabling better understanding of funding behavior.
- Logistic Regression and Random Forest models provide highly accurate predictions, showing that funding stage is strongly influenced by Investment Amount and Industry.
- K-Means clustering helps in segmenting startups into distinct groups (Low, Mid, and High Funded), which can be used for targeted investment strategies.
- Association rule mining reveals frequently occurring Industry-City-InvestmentType combinations, helping investors improve portfolio strategies.
- Combining classification, clustering, and association rules provides a comprehensive approach for startup ecosystem analysis and data-driven decision-making.

Observations

- Decision Tree achieved the highest accuracy among the three models used in this project.
- Logistic Regression and Random Forest also showed good classification performance.
- Precision, Recall, and F1-score values indicate balanced prediction capability across the models.
- Investment Amount was one of the most influential features affecting funding stage prediction.
- EdTech, Fintech, and HealthTech were among the most funded industries in the dataset.
- Bengaluru, Mumbai, and Delhi emerged as major startup funding hubs.

Conclusion

The project successfully applied multiple machine learning techniques to classify Indian startups based on their funding stage. Decision Tree showed the highest accuracy, while Logistic Regression and Random Forest also provided good prediction performance. Association rule mining revealed key industry-city patterns that drive startup funding in India. K-Means clustering helped segment startups into three distinct funding levels, providing actionable insights for investors and policymakers. Overall, the models demonstrate the effectiveness of data mining techniques in analyzing startup funding data and supporting data-driven investment decision-making.